Arithmetic obstructions in positive determinant approximations to $\beta(4)$

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Abstract

Small positive determinants offer a possible route to proving that $\beta(4) = 1 - 3^{-4} + 5^{-4} - \cdots$ is irrational. The determinant must be the value of an integer polynomial, and it must be sufficiently small relative to the polynomial's degree. We examine two explicit constructions and prove that clearing denominators necessarily destroys the required smallness, even after all common coefficient factors are removed. The resulting primitive integer-polynomial values grow at least exponentially in the square of the matrix size, with lower exponents $2 - 2\log 2$ and $6 - \log 4$. The first conclusion covers every consecutive block of positive integer nodes, regardless of how its starting point varies with the size. For the second construction, we determine the exact contribution of every sufficiently large prime to the primitive multiplier. These are failure theorems for the two specified families. We explain the irrationality criterion and the constructions before developing their positive integral representations, real estimates and prime-divisibility arguments.

1 Introduction

We study two attempts to prove that

$$\theta = \beta(4) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^4} \tag{1.1}$$

is irrational. Both attempts produce positive determinants that become very small. The question is whether these small numbers can be turned into a contradiction to rationality. We prove that, for these two constructions, the conversion to integer coefficients necessarily destroys the smallness. Theorems 1.1 and 1.2 are thus results about the limitations of particular proof strategies.

1.1 What would prove irrationality?

Suppose that $\theta = a/b$ were rational, with integers a, b and $b > 0$. For any integers A, B,

$$B\theta - A = \frac{Ba - Ab}{b}.$$

If this is nonzero, its numerator is a nonzero integer, so its absolute value is at least $1/b$. Consequently, a sequence of nonzero integer linear forms $B_n\theta - A_n$ tending to zero would prove irrationality: no fixed denominator b could accommodate the whole sequence.

This is stronger than finding accurate rational approximations. If A_n/B_n approximates θ, the relevant quantity is

$$|B_n\theta - A_n| = |B_n| \left|\theta - \frac{A_n}{B_n}\right|.$$

The approximation error must remain small after multiplication by its denominator. Every real number has arbitrarily accurate rational approximations; the integer-form requirement is what makes this an irrationality test.

1.2 Why use determinants?

If the entries in a matrix are rational linear forms in θ, its determinant is a polynomial with rational coefficients evaluated at θ. For example, with $r, s, t \in \mathbb{Q}$,

$$\det \begin{pmatrix} X + r & s \\ s & X + t \end{pmatrix} = X^2 + (r + t)X + rt - s^2.$$

An $n \times n$ matrix of linear entries similarly gives a polynomial of degree at most n, and exactly n in our two constructions. Increasing the dimension can help make the determinant small, but increasing the degree makes the irrationality test harder. We explain both effects.

On the real side, determinants can be especially small when their rows are almost linearly dependent. Orthogonalizing the rows of a real square matrix by subtracting combinations of earlier rows preserves the determinant. Its absolute value is the product of the lengths of the orthogonalized row vectors. Thus small components in successive new directions can multiply together. For illustration, lengths $e^{-c}, e^{-2c}, \ldots, e^{-nc}$, with $c > 0$, would give

$$|\det M_n| = \prod_{j=1}^{n} e^{-cj} = e^{-cn(n+1)/2}.$$

This explains how decay on the n^2 scale can arise. It is an illustrative mechanism; the rate for each actual construction requires proof, as does nonvanishing.

On the arithmetic side, the degree determines the lower bound under rationality. If $Z \in \mathbb{Z}[X]$ has degree at most n, then $b^n Z(a/b)$ is an integer. Thus

$$Z(\theta) \neq 0 \quad \Longrightarrow \quad |Z(\theta)| \geq b^{-n} \qquad \text{if } \theta = a/b. \tag{1.2}$$

Larger degree permits smaller nonzero values at rational arguments; this is a cost of the construction. The potential gain is that the determinant can shrink on the n^2 scale while the degree grows only linearly. If, after conversion to integer coefficients, its nonzero value is at most e^{-cn^2} for some fixed $c > 0$, then it is eventually smaller than b^{-n} for every fixed b, contradicting rationality. More generally, it suffices that $|Z_n(\theta)|^{1/n} \to 0$. Merely tending to zero is insufficient: $Z_n(X) = X^n$ has value 2^{-n} at the rational number $1/2$.

Positive determinant constructions are attractive because they can meet two requirements at once: rapid decay and nonvanishing. A *Gram matrix* consists of the pairwise inner products of a list of functions. If those functions are linearly independent, the matrix is positive definite and its determinant is positive, although it may be very small. Integral representations of this kind let us establish nonvanishing without knowing whether θ is rational. Zudilin [6] develops this approach systematically. The determinant strategy for the related value $\beta(2)$, Catalan's constant, described by OpenAI [4, Section 1.2], is a motivation for testing such constructions at $\beta(4)$.

1.3 The cost of integer coefficients

The determinant initially gives $Q_n \in \mathbb{Q}[X]$, whereas the test (1.2) requires integer coefficients. We therefore clear denominators and then divide out the greatest common divisor of all coefficients. The resulting integer polynomial is called *primitive*. Write

$$Z_n = \kappa_n Q_n = \mathrm{Prim}(Q_n), \qquad \log|Z_n(\theta)| = \log\kappa_n + \log|Q_n(\theta)|, \quad \kappa_n > 0. \tag{1.3}$$

Every integer polynomial proportional to Q_n is an integer multiple of Z_n. Thus Z_n has the smallest nonzero absolute value obtainable by rationally rescaling this same polynomial.

Removing common factors matters. For example, $Q(X) = (6X + 12)/5$ becomes integral after multiplication by 5, but its primitive representative is only $X + 2$: the actual multiplier is $5/6$. A large denominator chosen to clear the entries of a matrix could therefore be a poor estimate of the cost of its determinant. Brown [1, Section 7.4] discusses the role of common divisors and congruences in this arithmetic comparison.

To prove that a construction fails, we must bound the actual primitive value from below. An upper bound that fails to tend to zero establishes only that the estimate is inadequate. Our proofs instead show that the determinants cannot be smaller, or their primitive multipliers cheaper, by enough to make the construction work.

1.4 The two results and what they mean

The exact matrix entries are given in Sections 3.1 and 5.1. Here we describe the two designs and state the conclusions. Carreira [2] also investigates $\beta(4)$ determinant constructions and their primitive arithmetic. The results below give proofs of failure for the specified infinite families, including all common coefficient cancellation.

First construction: consecutive nodes. We build a symmetric matrix with rational off-diagonal entries and θ plus a rational on each diagonal entry. The same nodes are used for its rows and columns; this is the meaning of *principal*. We choose consecutive integer nodes

$$m + 1, m + 2, \ldots, m + n.$$

The parameter n is the matrix size and m is its offset. Replacing θ by X and taking the determinant defines $Q_{n,m}(X)$; write $Z_{n,m} = \mathrm{Prim}(Q_{n,m})$. The construction is made explicit in (3.2)–(3.3).

Theorem 1.1 (Consecutive principal determinants)**.** *For every sequence of nonnegative integer offsets m_n, $Q_{n,m_n}(\theta) > 0$ and*

$$\liminf_{n\to\infty} \frac{\log Z_{n,m_n}(\theta)}{n^2} \geq 2 - 2\log 2 > 0. \tag{1.4}$$

The positive constant $2 - 2\log 2$ is approximately 0.614. The inequality says that, for any chosen offset sequence m_n and any $\varepsilon > 0$, eventually

$$Z_{n,m_n}(\theta) \geq e^{(2-2\log 2-\varepsilon)n^2}.$$

These integer-polynomial values grow instead of becoming small. The offset may depend on n in any way, even growing much faster than n. Thus moving the consecutive block farther out cannot repair this family. The theorem concerns consecutive blocks, not arbitrary selections of nodes.

Second construction: moments of a positive weight. We use integrals of powers of $y = (u-v)^2$, with $u, v \in [0,1]$, against a positive weight. Such integrals are called *moments*. Write them as $\mu_0, \mu_1, \ldots$ and form the matrix $[\mu_{i+j}]_{0 \le i,j < n}$, the Gram matrix of $1, y, \ldots, y^{n-1}$. Each moment turns out to be a rational multiple of θ plus a rational. Replacing θ by X gives a determinant polynomial $Q_n^{\Delta}(X)$ and its primitive representative $Z_n^{\Delta} = \mathrm{Prim}(Q_n^{\Delta})$. Formula (5.4) gives their precise definitions; the superscript Δ denotes this difference-variable construction.

Theorem 1.2 (Difference-variable Gram determinants)**.** *The polynomial Q_n^{Δ} has degree n and leading coefficient 2^{n-1}, and $Q_n^{\Delta}(\theta) > 0$. Moreover,*

$$\log Z_n^{\Delta}(\theta) \ge (6 - \log 4)n^2 + o(n^2). \tag{1.5}$$

Here $o(n^2)$ denotes a term whose ratio to n^2 tends to zero. Since $6 - \log 4 \approx 4.614 > 0$, these primitive values also grow exponentially in n^2. In this family we can say more about the reason: Proposition 6.1 determines exactly how many powers of every prime $p > 4n$ remain in the primitive multiplier. That arithmetic result is stated alongside its proof in Section 6.

These conclusions go beyond observing that a list of examples fails. They rule out the required small values at all sufficiently large sizes, with every offset schedule included in the first family. They also account for all rational scalar rescalings, because the primitive polynomial already gives the smallest such value. The contribution is a precise explanation of failure for these two natural constructions. It does not decide whether θ is rational, or exclude different determinants, polynomial factors or combinations of the constructed expressions.

1.5 How the proofs work

The two proofs compare the gain from a small real determinant with the cost of making its coefficients primitive integers. Both use the same three steps:

1. Express the determinant through a positive integral, so its value cannot be zero.
2. Bound that value both above and below, determining its rate of decay. A lower bound is essential for a failure theorem.
3. Find prime powers that must occur in the primitive multiplier, even after every common coefficient factor has been removed.

The comparison is particularly simple at the leading quadratic scale. In the table, each entry is the coefficient of n^2 in the indicated logarithm; terms $o(n^2)$ are omitted. The first row uses offset $m = 0$.

Family	$\log Q_n(\theta)$	$\log \kappa_n$ (lower)	$\log Z_n(\theta)$ (lower)
Consecutive, $m = 0$	$-2\log 2$	2	$2 - 2\log 2$
Difference Gram	$-\log 4$	6	$6 - \log 4$

The negative second column describes the small determinants. Adding the arithmetic cost in the third column gives a positive fourth column: the best integer-coefficient values become large. The proof of Theorem 1.1 also checks that changing the offset cannot reverse this comparison.

Section 2 gives the arithmetic conventions. Sections 3–4 construct and analyze the first family; Sections 5–6 do so for the second. Each construction is introduced before its technical proof begins, and the two arguments can be read separately. Section 7 describes the scope and evidence; Appendix A includes a zero node in the first construction.

The introductory argument uses rational arithmetic and linear algebra. The detailed proofs also use calculus, elementary complex integration, Stirling's formula and the prime number theorem in the form

$$\vartheta(x) := \sum_{p \le x} \log p = x + o(x). \tag{1.6}$$

The latter counts the total logarithmic contribution of the primes whose powers the arithmetic arguments force. No numerical optimization or finite enumeration is an input to either asymptotic obstruction.

2 Primitive normalization and prime powers

Definition 2.1. For $0 \neq Q \in \mathbb{Q}[X]$, its primitive representative $\mathrm{Prim}(Q)$ is the unique positive rational multiple $\kappa Q \in \mathbb{Z}[X]$ whose coefficients have gcd one. This convention preserves the sign of Q.

We write $[X^j]Q$ for the coefficient of X^j in Q. For a nonzero rational number, v_p counts powers of p in its numerator minus those in its denominator; set $v_p(0) = +\infty$. For example $v_p(p^{-3}) = -3$. Saying that a rational number is p-integral means that its denominator in lowest terms is not divisible by p. Reduction modulo p below is always performed on p-integral quantities, by inverting the denominator modulo p. A p-unit has valuation zero, so its reduction is nonzero.

Lemma 2.2. *If $Q(X) = \sum_j q_j X^j \neq 0$ and $\mathrm{Prim}(Q) = \kappa Q$, then*

$$v_p(\kappa) = -\min_j v_p(q_j) \tag{2.1}$$

for every prime p. If Q is monic, κ is a positive integer. If the leading coefficient of Q is 2^e with $e \ge 0$, then $v_p(\kappa) \ge 0$ for odd p and $v_2(\kappa) \ge -e$.

Proof. Integral primitive coefficients have minimum valuation zero at every prime, which proves (2.1). A monic polynomial has a coefficient of valuation zero at every prime. A leading coefficient 2^e has valuation zero at odd primes and valuation e at 2. □

If $Z = \mathrm{Prim}(Q)$ and λQ has integer coefficients, then $(\lambda/\kappa)Z$ has integer coefficients. The denominator of λ/κ in lowest terms must divide every coefficient of Z, so it is one. Thus every integral rational rescaling is an integer multiple of Z, as asserted in the introduction.

Formula (2.1) tells us how to prove that a prime power is unavoidable: to show $v_p(\kappa) \ge r$, it suffices to find one coefficient of Q with valuation at most $-r$. To establish equality, we must also show that no coefficient has a lower valuation. The first construction uses the lower bound; the second obtains equality for every sufficiently large prime.

If $M(X)$ has rational polynomial entries, $\det M(X) \neq 0$, and U, V are invertible rational matrices of the same size, then $\det(UM(X)V) = \det U \det V \det M(X)$. Consequently the primitive representative of $\det M(X)$ is unchanged up to sign by rational invertible row and column operations. Passing to a different subspace or a different matrix is a different construction and is not covered by this observation.

3 First construction: a positive kernel

This section constructs the first family from a shifted beta-series tail. We then prove positivity and estimate the real determinant. Section 4 will determine the arithmetic cost.

3.1 Choosing the matrix entries

The first construction seeks a symmetric matrix with rational off-diagonal entries and θ plus a rational on each diagonal entry. We also want the matrix to be positive definite, so that its determinant cannot vanish. The following calculation explains the entries; the integral representation in the next subsection will establish positive definiteness.

For integers $a \geq 0$ and $\ell \geq 2$, write

$$S_\ell(a) = \sum_{j=0}^{a-1} \left(\frac{1}{(4j+1)^\ell} - \frac{1}{(4j+3)^\ell} \right), \qquad S_\ell(0) = 0. \tag{3.1}$$

These are the first $2a$ terms of the alternating beta series of order ℓ, grouped into pairs. In particular, $\theta - S_4(a)$ is the remaining fourth-power tail. To obtain it by differentiation, start from the shifted cubic tail

$$f(t) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1+4t)^3} \qquad (t \geq 0).$$

At a nonnegative integer a, shifting the summation index and differentiating give

$$f(0) - f(a) = S_3(a), \qquad f'(a) = -12\big(\theta - S_4(a)\big).$$

Thus subtracting the tail from $f(0)$ gives rational values at integer arguments, while its derivative introduces θ. Combine these properties by setting, for $x > 0$,

$$g(x) = \frac{\sqrt{x}}{6}\big(f(0) - f(\sqrt{x})\big), \qquad K(x,y) = \frac{g(y) - g(x)}{y - x}, \quad K(x,x) = g'(x).$$

Here K is a divided difference, with its limiting derivative on the diagonal. At squared integer nodes,

$$g(a^2) = \frac{aS_3(a)}{6}, \qquad g'(a^2) = \frac{S_3(a) - af'(a)}{12a} = \theta - S_4(a) + \frac{S_3(a)}{12a} \quad (a > 0).$$

The factor $\sqrt{x}$ compensates for the $1/(2\sqrt{x})$ from the chain rule, and the factor $1/6$ makes the coefficient of θ exactly one. These identities explain both the squared nodes and the constants in the matrix entries. For distinct positive integers $a_1, \ldots, a_n$, we have $[K(a_i^2, a_j^2)]_{i,j} = \theta I_n + R(a)$, where $R(a)$ is the rational symmetric matrix

$$R(a)_{ii} = -S_4(a_i) + \frac{S_3(a_i)}{12a_i}, \qquad R(a)_{ij} = \frac{a_j S_3(a_j) - a_i S_3(a_i)}{6(a_j^2 - a_i^2)} \quad (i \neq j). \tag{3.2}$$

The integral representation below proves that K is a positive kernel: its matrices at distinct positive nodes are positive definite. This does not follow from the divided-difference formula alone. Replacing θ by X gives a *matrix pencil* $XI_n + R(a)$, whose entries depend linearly on X and whose determinant is monic. For consecutive nodes $a_i = m + i$, $1 \leq i \leq n$, set

$$Q_{n,m}(X) = \det(XI_n + R(a)), \qquad Z_{n,m}(X) = H_{n,m} Q_{n,m}(X) = \mathrm{Prim}(Q_{n,m})(X), \tag{3.3}$$

where $H_{n,m} > 0$ and $Z_{n,m} \in \mathbb{Z}[X]$ has coefficient gcd one.

To obtain positivity and the size estimate, we express $K(x,y)$ as an integral of $(x+s^2)^{-1}(y+s^2)^{-1}$ against a positive density. These factors give a Gram representation when the row and column nodes agree. An explicit determinant integral handles different node lists. We then compare the density with simpler ones that yield Cauchy determinants.

3.2 A direct integral representation

Extend the preceding real-variable function to $\operatorname{Re} z \geq 0$ by

$$f(z) = \sum_{j=0}^{\infty} \left((4j+1+4z)^{-3} - (4j+3+4z)^{-3} \right). \tag{3.4}$$

Expanding $\operatorname{sech} u = 2\sum_{k\geq 0}(-1)^k e^{-(2k+1)u}$ and integrating gives

$$f(z) = \frac{1}{4}\int_0^{\infty} u^2 \operatorname{sech}(u) e^{-4zu}\,\mathrm{d}u, \quad f(a) = f(0) - S_3(a), \quad f'(a) = -12(\theta - S_4(a)) \tag{3.5}$$

at nonnegative integer a. The integrated series is absolutely summable, which justifies the expansion and differentiation used here.

The next two transforms convert the exponential in this Laplace integral into the rational factors needed for the kernel. The cosine transform evaluates the resulting density explicitly; the second identity replaces $\sqrt{x}(1 - e^{-t\sqrt{x}})$ by an integral containing $x/(x+s^2)$. We include their proofs to fix the normalization:

$$\int_0^{\infty} \operatorname{sech}(u)\cos(tu)\,\mathrm{d}u = \frac{\pi}{2}\operatorname{sech}\frac{\pi t}{2}, \tag{3.6}$$

$$\int_0^{\infty} \frac{x(1-\cos(ts))}{x+s^2}\,\mathrm{d}s = \frac{\pi\sqrt{x}}{2}(1 - e^{-t\sqrt{x}}), \qquad x > 0,\ t \geq 0. \tag{3.7}$$

For (3.6), integrate $e^{itz}/\cosh z$ around the rectangle with vertices $\pm R, \pm R + i\pi$. The top edge contributes $e^{-\pi t}$ times the bottom edge, and the pole at $i\pi/2$ has residue $e^{-\pi t/2}/i$. The vertical edges tend to zero. Hence the full-line transform is $\pi\operatorname{sech}(\pi t/2)$; symmetry gives the displayed formula. For (3.7), the upper-half-plane pole at $i\sqrt{x}$ gives $\int_0^{\infty}\cos(ts)/(s^2+x)\,\mathrm{d}s = \pi e^{-t\sqrt{x}}/(2\sqrt{x})$ for $t > 0$, and the value at $t = 0$ follows by continuity. Subtract the latter transforms and multiply by x.

Differentiating (3.6) twice under the integral gives

$$f(0) = \frac{\pi^3}{32}, \qquad \operatorname{Re} f(is) = \frac{\pi^3}{32}A(s)(2A(s)^2 - 1), \quad A(s) = \operatorname{sech}(2\pi s). \tag{3.8}$$

Define

$$g(x) = \frac{\sqrt{x}}{6}(f(0) - f(\sqrt{x})), \quad g(0) = 0, \qquad K(x,y) = \frac{g(y) - g(x)}{y - x}, \tag{3.9}$$

with the derivative value when $x = y$.

Proposition 3.1. *For $x, y \geq 0$,*

$$K(x,y) = \int_0^{\infty} \frac{\omega(s)}{(x+s^2)(y+s^2)}\,\mathrm{d}s, \quad \omega(s) = \frac{\pi^2}{96}s^2 q(s), \quad q(s) = 1 + A(s) - 2A(s)^3. \tag{3.10}$$

The density ω is strictly positive for $s > 0$. On squared integer nodes the kernel equals

$$K(a^2, b^2) = \frac{bS_3(b) - aS_3(a)}{6(b^2 - a^2)} \qquad (a \neq b), \tag{3.11}$$

$$K(a^2, a^2) = \theta - S_4(a) + \frac{S_3(a)}{12a} \qquad (a > 0), \tag{3.12}$$

$$K(0,0) = 2\theta. \tag{3.13}$$

Every square minor with strictly increasing nonnegative row nodes and strictly increasing nonnegative column nodes is positive.

Proof. First identify the density. Equations (3.5) and (3.7), with $t = 4u$, give

$$g(x) = \frac{1}{12\pi} \int_0^\infty \int_0^\infty u^2 \operatorname{sech} u \, \frac{x(1 - \cos(4us))}{x + s^2} \, \mathrm{d}s \, \mathrm{d}u.$$

The integrand is nonnegative, so Tonelli's theorem permits interchange. The inner u-integral is $4(f(0) - \operatorname{Re} f(is))$. Hence

$$g(x) = \int_0^\infty \frac{x}{x + s^2} \sigma(s) \, \mathrm{d}s, \qquad \sigma(s) = \frac{f(0) - \operatorname{Re} f(is)}{3\pi} = \frac{\pi^2}{96} q(s). \tag{3.14}$$

Taking a divided difference uses the elementary identity

$$\frac{y/(y + s^2) - x/(x + s^2)}{y - x} = \frac{s^2}{(x + s^2)(y + s^2)},$$

which gives (3.10). Positivity is now explicit:

$$q = (1 - A)(2A^2 + 2A + 1) > 0 \qquad (s > 0). \tag{3.15}$$

Since $\omega(s) = O(s^4)$ at zero and $O(s^2)$ at infinity, the integral is finite even when $x = y = 0$. Differentiation or dominated convergence justifies the diagonal and zero limits. The identities (3.11)–(3.12) follow from (3.5); at zero, $g(x) = -f'(0)x/6 + O(x^{3/2}) = 2\theta x + O(x^{3/2})$.

It remains to pass from the kernel to its minors. For increasing nodes x_i, y_j, expanding the two determinants and integrating termwise gives the determinant integration identity

$$\det[K(x_i, y_j)]_{i,j=1}^n = \frac{1}{n!} \int_{(0,\infty)^n} \det\left[\frac{1}{x_i + s_\ell^2}\right]_{i,\ell} \det\left[\frac{1}{y_j + s_\ell^2}\right]_{j,\ell} \prod_{\ell=1}^n \omega(s_\ell) \, \mathrm{d}s_1 \cdots \mathrm{d}s_n. \tag{3.16}$$

Absolute integrability follows by expanding into finitely many products of the finite one-variable integrals. This is the usual Andréief–Heine identity used in positive determinant arguments [6, Section 2], with its elementary expansion proof here. For $0 < s_1 < \cdots < s_n$, both Cauchy determinants are positive by the Cauchy product formula (3.17), applied to x_i, s_j^2 and y_i, s_j^2. Their product is unchanged by permuting the s_ℓ and is strictly positive away from coincidences. This proves the last assertion. It also proves that increasing the density increases these determinants, even for different row and column nodes. □

Remark 3.2. The density in (3.10) satisfies $\omega(s) \sim \pi^2 s^2/96$ at infinity; it has infinite total mass, but the two resolvent factors make every kernel entry integrable. Carreira's functional [2, Definition 24(B) and Lemma 25] uses a positive density obtained from derivatives of $1/\sinh(\pi s)$, with exponential decay and finite polynomial moments. The representation used here is derived directly, and assumes no polynomial moments of $\omega(s) \, \mathrm{d}s$.

3.3 A uniform comparison with Cauchy determinants

The positive representation makes nonvanishing transparent, but its density is inconvenient for estimating a large determinant. We compare that density with simpler ones whose determinants have explicit product formulas. The comparison changes the logarithm by only $O(n)$, so it preserves the n^2 scale relevant to the obstruction. Positivity of the determinant integrand is what makes this comparison legitimate.

For positive nodes a_i, b_j, write

$$C(a,b)_{ij} = \frac{1}{a_i + b_j}, \qquad \det C(a,b) = \frac{\prod_{i<j}(a_j - a_i)(b_j - b_i)}{\prod_{i,j}(a_i + b_j)}. \tag{3.17}$$

The product formula follows, for example, by multiplying the determinant by its displayed denominator: the numerator is alternating separately in the two node sets, hence contains their Vandermonde factors, and comparison of degrees and one residue fixes the remaining constant to one. The constants below have three roles: M bounds the density factor q, τ_0 fixes a lower comparison that vanishes quadratically at zero, and c_0 includes the factor $\pi/2$ from integrating the comparison kernel. Set

$$c_0 = \frac{\pi^3}{192}, \qquad M = 1 + \frac{2}{3\sqrt{6}}, \qquad \tau_0 = \frac{1}{\sqrt{2}\pi}. \tag{3.18}$$

Theorem 3.3 (Uniform Cauchy comparison)**.** *For any two strictly increasing lists of positive integers a_i, b_j,*

$$\left(\frac{c_0}{(1+\tau_0)^2}\right)^n \det C(a,b) \le \det[K(a_i^2, b_j^2)] \le (Mc_0)^n \det C(a,b). \tag{3.19}$$

In particular,

$$\log\det[K(a_i^2, b_j^2)] = \log\det C(a,b) + O(n), \tag{3.20}$$

with an absolute constant independent of all node positions and gaps.

Proof. The maximum of $1 + A - 2A^3$ for $0 \le A \le 1$ is M. Also $\cosh v \ge 1 + v^2/2$ and (3.15) imply

$$\frac{s^2}{s^2 + \tau_0^2} \le 1 - \operatorname{sech}(2\pi s) \le q(s) \le M. \tag{3.21}$$

The elementary rational integrals needed for the comparison are

$$\int_0^\infty \frac{s^2\,\mathrm{d}s}{(s^2+a^2)(s^2+b^2)} = \frac{\pi}{2(a+b)}, \tag{3.22}$$

$$\int_0^\infty \frac{s^4\,\mathrm{d}s}{(s^2+\tau^2)(s^2+a^2)(s^2+b^2)} = \frac{\pi}{2} R_\tau(a,b), \tag{3.23}$$

$$R_\tau(a,b) = \frac{ab + \tau(a+b)}{(a+b)(a+\tau)(b+\tau)}. \tag{3.24}$$

To check the second formula, first divide $s^4/((s^2+a^2)(s^2+b^2))$ into $[a^2 s^2/(s^2+a^2) - b^2 s^2/(s^2+b^2)]/(a^2-b^2)$, and then use the first formula with the extra factor $s^2+\tau^2$. Coincident parameters follow by continuity. The first formula follows by partial fractions and $\int_0^\infty (s^2+a^2)^{-1}\,\mathrm{d}s = \pi/(2a)$.

Applying the positive determinant integral (3.16) to the density bounds gives

$$c_0^n \det[R_{\tau_0}(a_i,b_j)] \le \det[K(a_i^2,b_j^2)] \le (Mc_0)^n \det C(a,b). \tag{3.25}$$

This uses the sign of the determinant integrand; an entrywise matrix comparison alone would not suffice.

We now evaluate the regularized determinant exactly:

$$\begin{aligned}\det[R_\tau(a_i,b_j)] = {} & \det C(a,b) \prod_i \frac{a_i}{a_i+\tau} \prod_j \frac{b_j}{b_j+\tau} \\ & \times \left(1 + \tau \sum_i \frac{1}{a_i} + \tau \sum_j \frac{1}{b_j}\right).\end{aligned} \tag{3.26}$$

Let $u_i = 1/a_i$, $v_j = 1/b_j$. The entrywise factorization

$$R_\tau(a_i, b_j) = \frac{a_i}{a_i + \tau}\left(\frac{1}{a_i + b_j} + \frac{\tau}{a_i b_j}\right)\frac{b_j}{b_j + \tau}$$

reduces the calculation to $\det(C + \tau uv^t)$. The rank-one identity $\det(C + \tau uv^t) = \det C(1 + \tau v^t C^{-1} u)$ leaves one scalar to compute. If $w = C^{-1}u$ and $F(x) = \sum_j w_j/(x + b_j)$, then $F(a_i) = 1/a_i$. Therefore

$$\frac{1}{x} - F(x) = \frac{1}{x}\,\frac{\prod_i (1 - x/a_i)}{\prod_j (1 + x/b_j)}.$$

Indeed, after multiplication by $x\prod_j(x + b_j)$, the numerator has degree at most n and vanishes at every a_i; its factor is fixed by the residue one at zero. Comparing constant terms at zero gives $v^t C^{-1} u = F(0) = \sum_i 1/a_i + \sum_j 1/b_j$. The rank-one determinant identity proves (3.26). Finally each integer node is at least one, so the products are at least $(1 + \tau_0)^{-2n}$ and the last factor is at least one. This proves (3.19) and (3.20). □

4 First construction: the arithmetic obstruction

We now determine part of the arithmetic cost in (1.3). For each prime in a suitable interval, the fourth-power tails contribute a diagonal residue that the cubic tails cannot cancel. A coefficient of the determinant records this residue. One such coefficient per prime is enough for a lower bound on the actual primitive multiplier.

4.1 A coefficient with forced valuation

For an odd prime p put $\chi_4(p) = (-1)^{(p-1)/2}$. Writing M_0 for the largest node, the interval $2M_0 < p < 4M_0$ below is chosen to isolate a single denominator p in each beta partial sum, while keeping all factors from differences of nodes invertible modulo p. Multiplication by p^4 then kills the cubic terms and detects only the fourth-power diagonal terms. We call a_i a high node when $4a_i > p$.

Lemma 4.1. *Let $a_1 < \cdots < a_n$ be positive integers with maximum M_0, and set $Q(X) = \det(XI_n + R(a))$. If $p \geq 5$ and $2M_0 < p < 4M_0$, put $r_p = \#\{i : 4a_i > p\}$. Then*

$$v_p([X^{n-r_p}]Q) = -4r_p, \qquad v_p(\kappa(Q)) \geq 4r_p, \tag{4.1}$$

where $\mathrm{Prim}(Q) = \kappa(Q)Q$.

Proof. The quantities a_i, $a_i + a_j$ and $a_i - a_j$ for $i \neq j$ are p-units. Among odd denominators below $4M_0$, the only possible multiple of p is p itself, since $3p > 4M_0$. It appears in $S_\ell(a_i)$ exactly when $4a_i > p$, with sign $\chi_4(p)$. Consequently $p^4 R(a)$ is p-integral and its reduction is diagonal:

$$p^4 R(a)_{ij} \equiv \begin{cases} -\chi_4(p), & i = j,\ 4a_i > p, \\ 0, & \text{otherwise} \end{cases} \pmod p. \tag{4.2}$$

Expanding $\det(XI + R)$ by the set of diagonal entries from which X is selected shows that the coefficient of X^{n-r_p} is the sum of all principal minors of R of size r_p. After multiplication by p^{4r_p}, exactly one such minor survives modulo p: that using all the high nodes. Its residue is $(-\chi_4(p))^{r_p} \neq 0$. For $r_p = 0$ this is the leading coefficient one. This proves the equality in (4.1); Lemma 2.2 proves the inequality for the actual primitive multiplier. □

Remark 4.2 (A three-node example). Take nodes $1, 2, 3$ and $p = 7$. The high nodes in the lemma are $2, 3$, and $\chi_4(7) = -1$, so

$$7^4 R \equiv \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \pmod 7.$$

The coefficient of X in $\det(XI_3 + R)$ is the sum of the three principal 2×2 minors:

$$[X]Q = (R_{11}R_{22} - R_{12}^2) + (R_{11}R_{33} - R_{13}^2) + (R_{22}R_{33} - R_{23}^2).$$

After multiplication by 7^8, only the final minor survives modulo 7, with residue 1. Thus $v_7([X]Q) = -8$, and the primitive multiplier must contain at least 7^8. This is the role of the uniquely surviving minor: it rules out cancellation in the sum defining that coefficient. Other coefficients may force additional powers; a lower bound is enough.

Since Q is monic, the primitive multiplier is integral. Thus the local inequalities can be summed without any negative contribution from omitted primes:

$$\log \kappa(Q) \geq 4 \sum_{2M_0 < p < 4M_0} r_p \log p \tag{4.3}$$

for M_0 sufficiently large. This is a lower bound after all coefficient content has been removed.

4.2 Proportional offsets

We now compare two functions of the relative offset m/n: $\mathcal{A}$ measures the forced denominator cost, and $\mathcal{D}$ measures the real decay. Their difference will be the lower bound for the primitive value. This subsection treats finite limiting offsets; the next handles $m/n \to \infty$ separately so that no uniformity in an unbounded parameter is assumed.

Take $a_i = m + i$, $M_0 = m + n$, and suppose first that $m/n \to \rho < \infty$. Reverse the finite sum in (4.3). Since the endpoints $2M_0$ and $4a_i$ are even and exceed two, endpoint conventions for primes have no effect. The prime number theorem, uniformly for $2M_0 \leq x \leq 4M_0$, gives

$$\begin{aligned} 4 \sum_{2M_0 < p < 4M_0} r_p \log p &= 4 \sum_{a_i > M_0/2} \left(\vartheta(4a_i) - \vartheta(2M_0)\right) \\ &= 16 \sum_{a_i > M_0/2} (a_i - M_0/2) + o(n^2) \\ &= \mathcal{A}(\rho) n^2 + o(n^2), \end{aligned} \tag{4.4}$$

where

$$\mathcal{A}(\rho) = \begin{cases} 2(1+\rho)^2, & 0 \leq \rho \leq 1, \\ 8\rho, & \rho \geq 1. \end{cases} \tag{4.5}$$

Reversing the prime sum counts, for each node a_i, the primes from $2M_0$ up to $4a_i$. With $a_i/n \to \rho + t$, the resulting Riemann sum is $16 \int_0^1 (\rho + t - (\rho+1)/2)_+ \, \mathrm{d}t$, where $u_+ = \max(u, 0)$. When $\rho \leq 1$, only the upper part of the node interval contributes; when $\rho \geq 1$, every node contributes. These are the two cases in (4.5). Uniformity of the prime-number-theorem error follows from $\sup_{2M_0 \leq x \leq 4M_0} |\vartheta(x) - x| = o(M_0)$ and $M_0 = O(n)$.

For the real estimate, the differences $a_j - a_i = j - i$ do not depend on the offset. The Vandermonde product is $\prod_{i<j}(j-i) = \prod_{j=1}^{n-1} j!$, so the Cauchy product is

$$\det C(a, a) = \frac{\prod_{j=1}^{n-1} (j!)^2}{\prod_{i,j=1}^{n} (2m + i + j)}. \tag{4.6}$$

Stirling's formula, followed by a Riemann sum for the denominator, gives

$$\log\det C(a,a) = -\mathcal{D}(\rho)n^2 + o(n^2), \qquad \mathcal{D}(\rho) = \frac{3}{2} + \int_0^1\int_0^1 \log(2\rho + x + y)\,\mathrm{d}x\,\mathrm{d}y. \tag{4.7}$$

For clarity, $2\sum_{j<n}\log(j!) = n^2\log n - \frac{3}{2}n^2 + O(n\log(n+1))$. At $\rho = 0$ the logarithmic corner singularity is integrable and its Riemann sums converge. Indeed, counting at most k grid points with $i + j = k$ bounds the contribution from $i + j \le \varepsilon n$ by $O(\varepsilon^2(1 + |\log\varepsilon|)) + O(\log(n+1)/n)$. A positive offset tending to zero is handled by monotonicity between offsets zero and any fixed $\varepsilon > 0$. Integrating explicitly yields

$$\begin{aligned}\mathcal{D}(\rho) = \frac{1}{2}\big((2\rho+2)^2\log(2\rho+2) - 2(2\rho+1)^2\log(2\rho+1) \\ + (2\rho)^2\log(2\rho)\big),\end{aligned} \tag{4.8}$$

where $0^2\log 0 = 0$.

It remains to show that shifting cannot make real decay overtake the arithmetic cost. Differentiating the integral formula shows that the derivative of $\mathcal{D}$ is decreasing, and

$$\mathcal{D}'(\rho) = 2\int_0^1\int_0^1 \frac{\mathrm{d}x\,\mathrm{d}y}{2\rho + x + y} \le \mathcal{D}'(0) = 4\log 2 < 4. \tag{4.9}$$

On both pieces of (4.5), $\mathcal{A}'(\rho) \ge 4$. It follows that

$$\mathcal{A}(\rho) - \mathcal{D}(\rho) \ge \mathcal{A}(0) - \mathcal{D}(0) = 2 - 2\log 2. \tag{4.10}$$

Combining (4.3), (4.4), (4.7) and Theorem 3.3 proves the stronger proportional-offset statement

$$\log Z_{n,m}(\theta) \ge (\mathcal{A}(\rho) - \mathcal{D}(\rho))n^2 + o(n^2). \tag{4.11}$$

4.3 Offsets larger than the matrix size

Suppose now that $m/n \to \infty$. All nodes exceed $M_0/2$ eventually. The relevant scale is now mn: the prime cost is of this order, whereas the logarithm of the real determinant has size at most $O(n^2\log(1 + m/n)) = o(mn)$. The same reversal of sums gives

$$\begin{aligned}\log H_{n,m} &\ge 16\sum_{i=1}^{n}(m + i - (m+n)/2) + o(nM_0) \\ &= 8mn + 8n + o(mn).\end{aligned} \tag{4.12}$$

The relative prime-number-theorem error is still uniform on $[2M_0, 4M_0]$, so this argument imposes no upper bound on m. By (4.6) and the numerator estimate just used,

$$\log\det C(a,a) \ge -n^2\big(\tfrac{3}{2} + \log(2(1 + m/n))\big) - O(n\log(n+1)) = -o(mn). \tag{4.13}$$

The Cauchy comparison costs only $O(n)$, whence

$$\log Z_{n,m}(\theta) \ge (8 - o(1))mn. \tag{4.14}$$

Every sequence of ratios m_n/n has subsequences converging in the compactified interval $[0,\infty]$. If the liminf asserted in Theorem 1.1 failed, a subsequence witnessing that failure would have a further subsequence with such a limit. Equations (4.11)–(4.14) exclude every possible limit. Positivity follows from Proposition 3.1. This completes the proof of Theorem 1.1.

5 Second construction: a moment determinant

The second design begins with a positive double integral. Its monomial integrals involve fourth-power beta tails on the diagonal and cubic tails off the diagonal. Expanding an even power of $u - v$ makes the unwanted $\beta(3)$ terms cancel, leaving rational linear forms in θ. Moreover $y = (u - v)^2$ lies in $[0, 1]$, so its powers form a positive moment sequence on a short interval. This gives both a nonvanishing Gram determinant and an effective estimate of its real size.

5.1 The moments and their determinant

For a polynomial $P(u, v)$, define the finite integral

$$J(P) = \int_0^1 \int_0^1 \frac{P(u,v)\log^2(uv)}{\sqrt{uv}(1+uv)}\,\mathrm{d}u\,\mathrm{d}v, \qquad \mu_m = \frac{1}{96}J((u-v)^{2m}). \tag{5.1}$$

The weight is integrable and strictly positive on the open square. Since $y = (u - v)^2$ lies in $[0, 1]$, the μ_m are moments of a positive measure on that interval. The matrix $[\mu_{i+j}]$ is the Gram matrix of $1, y, \ldots, y^{n-1}$; its determinant is positive because no nonzero polynomial vanishes throughout the interval. Positive-weight representations and Padé approximants for Hurwitz zeta values are developed by Prévost [5]. Rational modifications of moment functionals are part of the classical orthogonal-polynomial framework; Krattenthaler [3] gives determinant formulas including such modifications and repeated parameters.

To identify the polynomial in θ, expand $(u-v)^{2m}$. The middle term contributes $\binom{2m}{m}(\theta - T_4(m))$; paired off-diagonal terms have rational values because their cubic beta tails cancel. The factor $1/96$ normalizes $\mu_0 = \theta$. The calculation below proves $\mu_m = C_m\theta + B_m$, with the following rational coefficients:

$$T_\ell(j) = \sum_{k=0}^{j-1} \frac{(-1)^k}{(2k+1)^\ell}, \qquad C_m = \binom{2m}{m}, \tag{5.2}$$

$$B_m = -C_m T_4(m) + \frac{1}{6}\sum_{\substack{0\le j\le 2m\\ j\ne m}} \binom{2m}{j}\frac{T_3(j)}{j-m}, \tag{5.3}$$

$$Q_n^\Delta(X) = \det[C_{i+j}X + B_{i+j}]_{0\le i,j<n}, \qquad Z_n^\Delta = \kappa_n Q_n^\Delta = \mathrm{Prim}(Q_n^\Delta), \quad \kappa_n > 0. \tag{5.4}$$

Here $T_\ell(j)$ truncates the beta series after j individual terms, so $S_\ell(a) = T_\ell(2a)$. The central binomial coefficient C_m comes from the middle term just described; B_m collects the finite tail corrections and the off-diagonal terms. Empty sums are zero. In particular $B_0 = 0$ and $B_1 = -149/81$.

Lemma 5.1. *With C_m, B_m defined in (5.2)–(5.3),*

$$\mu_m = C_m\theta + B_m. \tag{5.5}$$

Proof. First compute the integral of one monomial; then expand $(u - v)^{2m}$. The identity

$$\int_0^1 \int_0^1 u^{a+t-1}v^{b+t-1}\,\mathrm{d}u\,\mathrm{d}v = \frac{1}{(a+t)(b+t)}$$

shows why two common-parameter derivatives insert $\log^2(uv)$. Expand $(1 + uv)^{-1}$ as a geometric series. For $a, b > 0$, the log-squared integral in each term is therefore the second derivative of the right side at $t = 0$. For $a \ne b$ this derivative is

$$\frac{2}{b-a}\left(\frac{1}{a^3} - \frac{1}{b^3}\right),$$

and for $a = b$ it is $6/a^4$. The sum of the absolute integrals of the terms is finite (their tails are $O(k^{-4})$), justifying termwise integration. Substitution of $a = k + i + 1/2$, $b = k + j + 1/2$ gives

$$\frac{J(u^i v^i)}{96} = (-1)^i(\theta - T_4(i)), \tag{5.6}$$

$$\frac{J(u^i v^j)}{96} = -\frac{(-1)^i}{6}\frac{T_3(i) - T_3(j)}{j - i} \quad (i \neq j,\ i + j \text{ even}). \tag{5.7}$$

In (5.7), the two cubic tails have the same sign $(-1)^i = (-1)^j$, so their $\beta(3)$ terms cancel. Now expand $\mu_m = \sum_{j=0}^{2m}(-1)^j\binom{2m}{j}J(u^j v^{2m-j})/96$. Its diagonal term contributes $\binom{2m}{m}(\theta - T_4(m))$ by (5.6). For $j \neq m$, the sign $(-1)^j$ cancels the corresponding sign in (5.7). The off-diagonal sum becomes

$$-\frac{1}{12}\sum_{\substack{0\le j\le 2m\\ j\neq m}}\binom{2m}{j}\frac{T_3(j) - T_3(2m - j)}{m - j}.$$

Changing j to $2m - j$ in the second part makes the two parts equal. The result is the sum in (5.3), proving (5.5). □

All coefficient statements use the explicitly defined polynomial $C_m X + B_m$. No independence or irrationality of θ is assumed to identify its coefficients.

5.2 Positive measure and full degree

On the half-square $u > v$, set $u = v + s$. The ranges are $0 < s < 1$ and $0 < v < 1 - s$. Symmetry doubles this integral, and then $y = s^2$ gives $2\,\mathrm{d}s = \mathrm{d}y/s$. Thus μ_m is the mth moment of a measure on $[0, 1]$ with density

$$\rho(y) = \frac{1}{96s}\int_0^{1-s}\frac{\log^2(v(v + s))}{\sqrt{v(v + s)}(1 + v(v + s))}\,\mathrm{d}v, \qquad 0 < y < 1. \tag{5.8}$$

This density is positive throughout $(0, 1)$ and has total mass $\mu_0 = \theta$. Consequently

$$D_n := Q_n^{\Delta}(\theta) = \det[\mu_{i+j}]_{0\le i,j<n} > 0 \tag{5.9}$$

for every n: a nonzero polynomial has positive squared norm.

Lemma 5.2. *For $n \geq 1$,*

$$\det[C_{i+j}]_{0\le i,j<n} = 2^{n-1}. \tag{5.10}$$

Thus Q_n^{Δ} has exact degree n and leading coefficient 2^{n-1}.

Proof. The C_m are the moments of the probability arcsine measure $\mathrm{d}y/(\pi\sqrt{y(4 - y)})$ on $[0, 4]$. This follows either by the beta integral or by the substitution $y = 2 + 2\cos t$, $0 < t < \pi$, and the binomial expansion of $(2 + 2\cos t)^m$. Let T_j denote the Chebyshev polynomial satisfying $\mathsf{T}_j(\cos t) = \cos(jt)$. The monic integer polynomials

$$c_0(y) = 1, \qquad c_j(y) = 2\mathsf{T}_j((y - 2)/2) \quad (j \geq 1)$$

are orthogonal for this measure, with squared norms $1, 2, 2, \ldots$. Monicity and integrality follow from $c_1 = y - 2$, $c_2 = (y - 2)^2 - 2$ and the Chebyshev recurrence for $j \geq 2$. The coefficient change from monomials has determinant one. The moment determinant is therefore the product of these norms, giving (5.10). □

5.3 Matching upper and lower real exponents

The upper estimate needs monic polynomials uniformly small on $[0,1]$; Chebyshev polynomials supply them. For the lower estimate we compare the measure with $(1-y)\,\mathrm{d}y$, whose monic orthogonal norms can be computed exactly. Changing from monomials to any monic polynomial basis uses a triangular matrix with determinant one, so it preserves the Gram determinant in both calculations.

Proposition 5.3. *For all $n \geq 1$,*

$$\frac{1}{6144^n n!\, 2^{2n^2+n}} \leq D_n \leq \theta^n 4^{-(n-1)^2}. \tag{5.11}$$

In particular,

$$\log D_n = -(\log 4)n^2 + O(n\log(n+1)). \tag{5.12}$$

Proof. For $j \geq 1$ the monic shifted Chebyshev polynomial on $[0,1]$ has supremum norm $2/4^j$. Change the monomial basis of the Gram matrix to these monic polynomials and apply Hadamard's inequality. Since the measure has total mass θ,

$$D_n \leq \theta^n \prod_{j=1}^{n-1} (2/4^j)^2 = \theta^n 4^{-(n-1)^2}.$$

For the lower bound restrict the integral in (5.8) to $v \in [(1-s)/4, (1-s)/2]$. There $v(v+s) \leq 1/2$, so $-\log(v(v+s)) \geq \log 2 \geq 1/2$, and the denominator of the integrand is at most 2. The integrand is at least $1/8$, giving

$$\rho(y) \geq \frac{1-s}{3072s} \geq \frac{1-y}{6144} \quad (0 < s < 1). \tag{5.13}$$

The second inequality is $2 \geq s + s^2$. Ordering the positive definite Gram matrices therefore gives

$$D_n \geq 6144^{-n} \det\left[\frac{1}{(i+j+1)(i+j+2)}\right]_{0\leq i,j<n} = 6144^{-n} \prod_{j=0}^{n-1} \frac{1}{(2j+2)\binom{2j+1}{j}^2}. \tag{5.14}$$

Here determinant monotonicity follows by conjugating the matrix inequality by the inverse square root of the smaller positive matrix.

We include the norm calculation in the last equality. For weight $1-y$ on $[0,1]$, the monic orthogonal polynomial of degree j is

$$p_j(y) = \frac{(-1)^j (j+1)!}{(2j+1)!(1-y)} \frac{\mathrm{d}^j}{\mathrm{d}y^j}(y^j(1-y)^{j+1}).$$

The numerator is divisible by $1-y$. Integration by parts j times shows orthogonality to degrees below j; every boundary term vanishes. The same calculation with p_j gives its squared norm

$$\frac{j!(j+1)!}{(2j+1)!}\int_0^1 y^j(1-y)^{j+1}\,\mathrm{d}y = \frac{j!^2(j+1)!^2}{(2j+1)!(2j+2)!} = \frac{1}{(2j+2)\binom{2j+1}{j}^2}.$$

The Gram determinant is the product of these monic norms. Finally $\binom{2j+1}{j} \leq 2^{2j+1}$ in (5.14) gives the lower bound in (5.11). Taking logarithms, using $0 < \theta < 1$ and Stirling's formula, proves (5.12). □

6 Second construction: the arithmetic obstruction

Here the prime-denominator contributions first appear at a specific moment index. Because a moment matrix has entry index $i+j$, their residues occupy a corner block with a nonzero anti-diagonal. Its determinant supplies the noncancelling coefficient, just as the single surviving minor did in Remark 4.2. In this family we also bound every coefficient's denominator at the same prime, which turns a lower bound into an exact formula.

Fix $n \geq 1$ and a prime $p > 4n$. All moment indices used in Q_n^Δ satisfy $0 \leq m \leq 2n-2$. In particular $2m < p$, $|j-m| < p$ for the indices in (5.3), and $T_4(m)$ is p-integral. In $T_3(j)$ the only possible odd denominator divisible by p is p itself: the largest odd denominator is at most $8n-9 < 2p$, whereas the next odd multiple is $3p$. It appears when $j \geq (p+1)/2$.

6.1 The first residue and its block

It follows that $p^3 B_m$ is p-integral and

$$p^3 B_m \equiv \frac{\chi_4(p)}{6} \sum_{j=h}^{2m} \binom{2m}{j} \frac{1}{j-m} \pmod{p}, \qquad h = \frac{p+1}{2}. \tag{6.1}$$

An empty sum is zero; in the range of this sum $j > m$, so no zero denominator occurs. Put

$$m_0 = \left\lfloor \frac{p}{4} \right\rfloor + 1, \qquad r = \max\{0, 2n-1-m_0\}. \tag{6.2}$$

Here m_0 is the first moment index for which the sum in (6.1) can be nonempty: $2m \geq (p+1)/2$ is equivalent to $m \geq \lfloor p/4 \rfloor + 1$. For $m < m_0$ no cubic tail contains p, so B_m is actually p-integral. If $m_0 \leq 2n-2$, the first nonzero residue in (6.1), apart from its factor $\chi_4(p)/6$, is

$$\begin{cases} 1/m_0, & p = 4m_0 - 1, \\ (2m_0-1)(m_0+1)/(m_0(m_0-1)), & p = 4m_0 - 3. \end{cases} \tag{6.3}$$

In the first case the sum has only $j = 2m_0$; in the second it has $j = 2m_0 - 1, 2m_0$, giving $2m_0/(m_0-1) + 1/m_0$. These residues are nonzero modulo p. Indeed, in the nonempty case $m_0 \geq 3$, and all the displayed positive numerator and denominator factors are less than p.

If $r = 0$, every residue vanishes. Otherwise, row i can contain a nonzero residue only if $i+j \geq m_0$ for some $j \leq n-1$, hence $i \geq m_0 - (n-1) = n - r$; the same applies to columns. The residue matrix $[p^3 B_{i+j} \bmod p]_{i,j<n}$ is therefore supported in the bottom-right $r \times r$ block. That block is anti-triangular with the nonzero first residue on its anti-diagonal, so it has rank r. More strongly, every entry outside it is p-integral before multiplication by p^3: its moment index is below m_0.

6.2 The exact primitive exponent

Proposition 6.1 (Exact large-prime cost)**.** *For every $n \geq 1$ and every prime $p > 4n$, the primitive multiplier κ_n of Q_n^Δ satisfies*

$$v_p(\kappa_n) = 3 \max\left\{0,\, 2n - 2 - \left\lfloor \frac{p}{4} \right\rfloor \right\}. \tag{6.4}$$

Proof of (6.4). At most r entries of a permutation term can have negative valuation, because only the bottom r rows have such entries. Each entry has coefficient valuations at least -3, so every coefficient of Q_n^Δ has valuation at least $-3r$.

To see that the bound is attained, suppose $r > 0$, let L be the $r \times r$ residue block just described, and put $s = n - r$. Multiply the bottom r rows of the pencil by p^3. The new matrix has p-integral polynomial entries, and its reduction has the form

$$\begin{pmatrix} [C_{i+j}X + B_{i+j}]_{0\le i,j<s} & * \\ 0 & L \end{pmatrix}.$$

The lower-left block vanishes because those entries were already p-integral. In the lower-right block the integral X-coefficients also vanish after multiplication by p^3. Taking this block-triangular determinant proves the polynomial congruence

$$p^{3r} Q_n^{\Delta}(X) \equiv \det(L) \det[C_{i+j}X + B_{i+j}]_{0\le i,j<s} \pmod{p}. \tag{6.5}$$

The right-hand coefficient of X^s is $\det(L)2^{s-1}$ by Lemma 5.2, and is a p-unit. Here $s \ge 1$, since $p > 4n$. Thus the minimum coefficient valuation is exactly $-3r$. If $r = 0$, all entries are p-integral and the leading coefficient 2^{n-1} is already a p-unit. Lemma 2.2 now proves (6.4) in all cases. □

This argument controls the actual primitive multiplier. Any coefficient gcd is already included in the minimum-valuation calculation; it cannot remove the forced exponent.

6.3 Summing the prime cost

By Lemma 2.2 and the leading coefficient 2^{n-1}, $v_p(\kappa_n) \ge 0$ at every odd prime, while $v_2(\kappa_n) \ge -(n-1)$. Hence

$$\log \kappa_n \ge 3 \sum_{4n<p<8n-8} \left(2n - 2 - \left\lfloor \frac{p}{4} \right\rfloor\right) \log p - (n-1)\log 2. \tag{6.6}$$

Changing the endpoints by a bounded amount and removing floors costs $O(n \log(n+1))$, which is negligible on the quadratic scale. Partial summation and (1.6) give

$$\begin{aligned} 3 \sum_{4n<p<8n-8} \left(2n - 2 - \left\lfloor \frac{p}{4} \right\rfloor\right) \log p &= 3 \int_{4n}^{8n} (2n - t/4)\, \mathrm{d}\vartheta(t) + O(n \log(n+1)) \\ &= 3 \int_{4n}^{8n} (2n - t/4)\, \mathrm{d}t + o(n^2) = 6n^2 + o(n^2). \end{aligned} \tag{6.7}$$

For the replacement of $\mathrm{d}\vartheta$ by $\mathrm{d}t$, integration by parts bounds the error using $\sup_{4n\le t\le 8n} |\vartheta(t) - t| = o(n)$, the $O(n)$ endpoint weight and its bounded derivative. Thus $\log \kappa_n \ge 6n^2 + o(n^2)$. Combining with Proposition 5.3 proves (1.5) and completes the proof of Theorem 1.2.

7 Scope, evidence and the wider investigation

The two proofs establish actual primitive-value obstructions. The first uses a monic pencil and a unique surviving principal minor; the second uses a power-of-two leading coefficient and a full anti-triangular residue block. In both cases the leading coefficient controls possible negative valuations of the primitive multiplier at the primes not used in the lower-bound sum. Without that control, a large denominator cost alone would not prove the same conclusion.

These results concern the full pencils defined in (3.3) and (5.4). Invertible rational basis changes preserve their primitive polynomials, but different subspaces, sparse designs, nonprincipal minors, polynomial weights, factors and combinations of determinants are outside the two obstruction

theorems. The uniform real comparison in Theorem 3.3 does apply to nonprincipal designs; an arithmetic obstruction for such a design would require additional information.

The accompanying research catalog separates proved limitations of defined families from failed estimates, exact reductions to existing constructions, finite experiments and incomplete constructions. In particular, some nonprincipal designs have unfavorable proved denominator bounds without an all-index bound on numerator content. An ordinary-moment first-variation construction has a positive error and a proved real exponent, but no all-index primitive arithmetic conclusion. Neither status is upgraded to an impossibility statement by the present theorems. The accompanying technical supplement develops selected further proofs and guides the reader through the remaining constructions, with links to the catalog and source snapshots.

Computational checks and authorship

This note was prepared with Codex. The archived record includes model mathematical cross-checks, exact rational calculations and Arb interval calculations. A reproducibility record accompanies this draft and states which checks were rerun and how the archived outputs were preserved. Such checks are finite evidence for identities and arithmetic formulas; the asymptotic results are proved in the text. No proof-assistant verification or human mathematical audit is claimed. The comparison with prior work is recorded by source and version; no exhaustive priority claim is made.

A Including a common zero node

The unregularized Cauchy matrix is unsuitable when both node lists contain zero, but the regularized matrix in (3.24) remains finite. For $a_0 = b_0 = 0$ and increasing positive remaining nodes, write a_+, b_+ for the lists with zero removed. Continuity gives $R_\tau(0,0) = 1/\tau$ and

$$\det[R_\tau(a_i, b_j)]_{0\le i,j<n} = \frac{1}{\tau}\prod_{i>0}\frac{a_i}{a_i+\tau}\prod_{j>0}\frac{b_j}{b_j+\tau}\det C(a_+, b_+). \tag{A.1}$$

To prove it, multiply row i by $a_i + \tau$ and column j by $b_j + \tau$. The resulting entry is $\tau + ab/(a+b)$, with continuous value τ in the zero row and column. Subtract the zero row from every other row and expand at the zero column. Undoing the scalings gives (A.1); an empty determinant is one.

The lower density bound (3.21) still applies. For an upper bound finite at zero, note that $1 - \operatorname{sech} v \le v^2/2$ for $v \ge 0$: integrate $\operatorname{sech} v \tanh v \le v$. Therefore

$$q(s) \le 5(1 - \operatorname{sech}(2\pi s)) \le 10\pi^2 s^2, \qquad q(s) \le M.$$

Combining the bounds gives $q(s) \le (M + 10\pi^2\tau^2)s^2/(s^2+\tau^2)$ for every $\tau > 0$. Choose $\tau = \tau_0$; then

$$c_0^n \det R_{\tau_0} \le \det[K(a_i^2, b_j^2)] \le [c_0(M+5)]^n \det R_{\tau_0}. \tag{A.2}$$

For positive integer remaining nodes, (A.1) gives

$$\log\det[K(a_i^2, b_j^2)] = \log\det C(a_+, b_+) + O(n), \tag{A.3}$$

uniformly in the remaining node lists.

Corollary A.1. *For the principal nodes $0, 1, \dots, n-1$, replace θ by X in (3.11)–(3.13), take the determinant, and let Z_n^0 be its primitive representative. Then*

$$\liminf_{n\to\infty}\frac{\log Z_n^0(\theta)}{n^2} \ge 2 - 2\log 2.$$

Proof. The coefficient matrix of X is $\operatorname{diag}(2, 1, \ldots, 1)$, so the leading coefficient of the determinant is 2. For primes $2(n-1) < p < 4(n-1)$, the unique high-node minor from Lemma 4.1 is unchanged; its complementary diagonal coefficient contributes a factor 2, a p-unit. Thus the same coefficient valuation $-4r_p$ is attained. All odd prime valuations of the primitive multiplier are nonnegative, and the only possible negative contribution is at least $-\log 2$. The prime sum is $2(n-1)^2 + o(n^2)$. By (A.3), its positive real determinant has logarithm $-2\log 2\,(n-1)^2 + o(n^2)$. Adding proves the assertion. □

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